Robert Leggieri

APPLIED AI & NLP DEVELOPMENT

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PROFESSIONAL SUMMARY

M.S. in Artificial Intelligence with project work in transformer training, NLP robustness, and reproducible model evaluation. Develops Python tools that connect data, experiments, and inspectable outputs. Brings more than twenty years of professional technology operations experience; seeking applied NLP and model development roles alongside researchers and software engineers.

TECHNICAL SKILLS

Model development: Python, PyTorch, Hugging Face Transformers, scikit-learn, LoRA / QLoRA

Data / evaluation: NumPy, pandas, SQL / SQLite, grouped evaluation, calibration, pytest, Git

Applications / integration: REST APIs, ETL, React, Node.js, Socket.IO, PowerShell

SELECTED PROJECTS

NLP Robustness through Training-Data Analysis

Graduate NLP project, UT Austin | Paper

- Instrumented DeBERTa training across 392,702 MNLI examples over five epochs; implemented category-weighted cross-entropy with a custom Hugging Face callback, weighted dataset, and Trainer override.
- Selected weighting improved reported HANS accuracy from 77.49% to 78.59% (+1.10 points), alongside MNLI accuracy of 88.91% to 88.47% (-0.44 points); documented selection and uncertainty limits.

Auditable Data Integration for Planning and Reporting

Professional project, Gutschick, Little & Weber, P.A. | GitHub

- Built a read-only BQE CORE connector that normalizes authorized operational and financial data into a validated SQLite warehouse and traceable outputs for an internal planning workflow.
- Implemented OAuth PKCE, permission-aware incremental sync, retries, deduplication, validation, and synthetic demonstrations.

Reproducible Text / Speech Model Evaluation

Public alpha reference software, released July 2026 | GitHub

- Developed dataset-neutral interfaces for leakage-aware grouping, nested evaluation, calibration, reliability, bootstrap intervals, and evidence provenance.
- Published bounded text-component evidence; explicitly separated audio runtime checks and synthetic pipeline verification from scientific validation.

PraxiCom: Faculty-Directed Voice Simulation

Archived spring 2025 academic prototype, UT Austin NURSING-AI Challenge | GitHub

- Built React student and faculty interfaces, a Node.js / Socket.IO relay, synthetic personas, and faculty evaluation around Hume EVI, with guidance from nursing faculty.

PROFESSIONAL EXPERIENCE

Network Manager - Gutschick, Little & Weber, P.A.

Burtonsville, MD | July 2004 - Present

- Use Python and PowerShell to automate workflows, integrate business systems, and translate requirements into dependable, secure technical solutions.
- Own technology operations, security, vendors, and infrastructure planning for an engineering firm supporting more than 40 users.

EDUCATION

The University of Texas at Austin - M.S., Artificial Intelligence, GPA 4.0 | August 2026

University of Maryland Global Campus - B.S., Data Science, Summa Cum Laude | May 2024